

## Morphicity: What the Action Is

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There are polymorphic actions and mimeomorphic actions. Mimeomorphic actions are actions that we either seek to *or are content to* carry out in pretty much the same way, in terms of behavior, on different occasions. Polimorphic actions are all the rest.<sup>1</sup>

For our purpose, we are going to use the term “behavior” in a special sense. In ordinary language, as well as in the philosophical literature, behavior is roughly equivalent to “internally produced movement or change.”<sup>2</sup> Understood in this way, human actions are a species of behavior, other species of behavior being the behavior of machines, animals, or plants. It is more useful in this context to define the notion of “behavior” as any sequence of bodily movements. Behaviors are what observers see people doing; they can be described by mapping bodily movements onto space–time coordinates.<sup>3</sup> Actions, on the other hand

1. In some sense we are engaged in the phenomenology of action, which is a step beyond the realm of issues traditionally addressed by the philosophy of action. The dominant tradition in the philosophy of action has been concerned with issues such as the logical structure of action sentences, freedom of the will, the nature of intentions, the causal or noncausal relation between intentions and actions, the differences between actions and happenings, and the viability of “folk theory” in describing, explaining, and predicting human actions. The almost exclusive focus on these problems has left other philosophical questions concerning actions underanalyzed. We are thinking here of the tradition in which the works of Anscombe (1979), Chisholm (1970), Danto (1973), Davidson (1982), and Goldman (1970) are seminal.

2. Cf. Dretske (1991), p. 3.

3. Our definition of behavior does not obviously cover “behaviors” such as calculating “in one’s head” where no bodily movements are visible. Nevertheless, as has been shown elsewhere (Collins 1990), such “mental” actions can be

are, always associated in one way or another with an intention (even though the actor may not be consciously aware of the intention from moment to moment). Therefore, for us, an action is the behavior plus the intention. An outside observer may not always see the difference between a piece of behavior that is the counterpart of an action and one that is not, but it is clear that it makes sense to talk of such a difference. Of course, in the way we use the term, action also involves a relationship to a form of life.

We can now say something about the difference between our two generic types of actions—polimorphic actions and mimeomorphic actions.

### Polimorphic Actions

Each of the generic types of action has subspecies. The simplest way to understand polimorphic actions is to note that in the case of many types of actions, the *same action* can be carried out by an indefinite number of *different behaviors*. For example, paying money can be done by passing metal or paper tokens, writing a check, offering a plastic card and signing a chit, and so forth, and each of these can be done in an open-ended variety of ways. At the same time, the *same piece of behavior* may be the instantiation of many *different actions*. For example, signing one's name might be the action of paying money, or it might be agreeing to a divorce, the final flourish of a suicide note, or a specimen signature for the bank.

Polimorphic actions are “rule bound” in the sense that it is usually clear when they are being done the wrong way. For example, there are wrong ways of doing even so ill-defined a thing as going for a walk. One

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understood as being associated with what can be thought of as internalized behaviors. Thus if one thinks of a child's action of adding (on the fingers) as being the coordination of chanting and finger movements, we may think of mental arithmetic as being the same thing done in “the mind's eye.” The analysis of both “mental actions” and actions that involve no movement, such as pretending to be a statue, is the same, but to sort out their conceptual structure it is easier to think about actions that do have a counterpart in observable behavior. (One of the reasons we use the term “behavior” rather than “movement” is because a subset of the actions that we analyze involve no intentional movement.)

way to do it wrong in many societies would be to brush against others on the sidewalk. Though polymorphic actions are rule bound, it is not possible to provide a recipe for doing them the right way that could be followed by someone who did not already understand the society in which they are embedded—there are too many context-dependent possibilities.

All types of (formative intentional) actions that are *not* mimeomorphic actions are “polimorphic actions.” The presence of variability in behavioral instantiation is characteristic of polymorphic actions. The variability with which an action of type “a” can be realized is thus part of what the action of type “a” means for the agent. Polimorphic actions are characterized by essential variability in the behaviors with which they are executed. Polimorphic actions are characterized by the fact that they usually involve varying behavior to carry out the same action in relation to a situation. The prefix “poly” connotes “manyness,” referring to the many behaviors that must correspond to a polymorphic action, but we have used the pun “poli” to connote that the appropriate behavioral shape of such an action has to be determined by reference to the society (polis). Thus, polymorphic actions are both many-shaped and take their shape from society.

If we separate the variations in situation from the choice of behavioral responses, we can distinguish four types of actions, three of which are polymorphic. There are *open polymorphic actions* in which both situation and response are open; there are *occasioned polymorphic actions* in which only the situation is open; and there are *playful polymorphic actions* in which only the behavioral response is open. In the fourth class of actions both situation and behavioral response are determinate; these are *mimeomorphic actions*. We now look briefly at each type of polymorphic action.<sup>4</sup>

### Open Polimorphic Actions: Love Letter Writing

We take writing love letters as an example of an *open polymorphic action*. Writing a love letter is an action such that different ways of carrying out

4. We adopt the convention of italicizing the first references to subspecies of action in each section, since each subspecies consists of three or four words. The italics distinguish subspecies from qualifying adjectives.

the action are not only possible but desirable. In fact, the difficulty (for some writers) of writing love letters over a prolonged period of time (and to the same beloved person), is to invent ever new ways of expressing one's love and affection that are still meaningful. A love letter that was an exact reproduction of an earlier one would be considered an insult or joke. Of course we can imagine circumstances under which an exact reproduction would still function as a love letter—say, where the beloved asked for such reproduction as a token of love—but these are special circumstances. Usually, writing a love letter will be a highly contextualized action, relating to the stylistic conventions of the times (courtly love, the restrained formality of Victorian society, the new freedoms of the 1960s), the form of life of the participants (gentlepersons, the Amish, punks), and the setting of the letters within the series of earlier letters and the major happenings in the world (periods of prosperity, major natural disasters, wars).

Though love letter writers are expected to be creative, it would be a misunderstanding to take an “open” choice to mean an “unbounded” choice. The creativity is bounded by conventions; there is a difference between a love letter and, say, a business letter. But these conventions—the conventions that keep society ordered—are conventions that constrain *actions*. Conventions that constrain *actions* put nothing like same the constraint on *behaviors* (as we use the term). To repeat the crucial point, to remain within the convention of the action “love letter writing” requires creativity in behavior. Love letter writing, then, requires an imaginative response to open-ended, changing circumstances, and it requires creativity within the bounds of convention; both context and response are open.

### Occasioned Polimorphic Actions: Voting

A good example of an *occasioned polimorphic action* is voting. Imagine a system in which members of parliament are asked to vote on bills that come before the house by pressing one of three buttons. The three buttons signify “for,” “against,” and “abstain.” Imagine, if possible, that these parliamentarians vote with their conscience every time. The members' behavioral repertoire is severely constrained in this case—they can do one of only three things—but they respond with one of these three

behaviors to an open ended environment consisting of the bills that come before them and the ever-changing world to which the parliamentary debates are a response or a contribution. An outside observer who did not understand the nature of a parliament or the world in which it was set would be unable to predict members' voting patterns even though there were only three possibilities. Thus even the identical behavior—say, pressing the “for” button—might comprise entirely different actions each time. Pressing the “for” button might be voting to, say, legalize abortion, to outlaw abortion, or to raise taxes on chocolate.

To summarize, *occasioned polymorphic actions* happen when a limited set of behaviors is used to respond in a meaningful way to an open context.<sup>5</sup>

### **Playful Polimorphic Actions: Relieving Tedium, Expressing Individuality**

A typical example of a *playful polymorphic action* occurs when assembly-line workers vary the way they manage a repetitive tedious job, not as a meaningful response to an open environment, and not because variation is needed to do the job properly, but because they are trying to relieve the tedium of the task without loss of efficiency. Behavioral variations of this sort might well be random; they are not socially embedded.<sup>6</sup>

5. When one types a letter, one has a limited choice of keys to press. Therefore, for all letters of finite length, one is choosing from what one might think of as a finite (albeit very large) choice of combinations of key presses. Why, then, are not all typed letters instances of occasioned polymorphic actions like the voting case where the choice was limited to one of a set of button pressings? The answer is that while all button pressings are potential votings, not all combinations of key presses are potential letter writings—in fact most are not: they are just nonsense. To understand which of the set of all combinations of key presses comprise letter writings, one has first to understand the society in which letter writing is taking place, and that is why the behavioral choice is not predetermined as it is in the case of voting by pressing buttons.

The fact that each of the “terminal actions” involved in writing a letter—pressing a key—comprises one choice from a limited range does not affect the argument, for key pressing is not letter writing. Key pressing is an action embedded within the action of letter writing.

6. Though in practice, playfulness at work is probably more ritualized than random, it does not effect the conceptual distinction we are making. There may be few pure examples of the types of actions we describe to be found in the world

Playful polymorphic actions fill up the logical space we have created by separating behavioral responses from the context in which they take place. It is characterized by a limited context and an open-ended repertoire of behaviors.

A variant of playful polymorphic actions is personalized action. An example would be the deliberate use of handwriting, as opposed to typing, in the production of a letter. In writing by hand, the writer introduces variation into symbol formation so as to reveal that the origin of the text was not a machine, but the variations are not meaningful. Here the variation from letter to letter and word to word is a way of symbolically expressing individuality, care, and personal effort.

While then, such actions are responses to social situations (whereas the previous example of the bored production-line worker was a response to an individual, psychological need), the actual variation in the formation of the symbols is not socially meaningful—that there is variation is meaningful, but the particular variations are not. Playful polymorphic actions, in their personalized variant, are formative actions, because their meaning is rooted in our forms of life, but they are, paradoxically, an institutionalized way of expressing individuality.<sup>7</sup>

Figure 3.1 shows the relationship of the three types of polymorphic actions and of mimeomorphic actions.

### **Mimeomorphic Actions**

The starting point for understanding mimeomorphic actions is to think of them as actions where, by contrast with polymorphic actions, we attempt to maintain a one-to-one mapping between our actions and observable behaviors. As in the execution of all actions, we may intend to execute mimeomorphic actions in a certain way, but this is not necessarily what we accomplish. Nevertheless, the proper description of the type of action is informed by our intention to do things in the same way

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as we know it. The point is, however, to describe exhaustively possible types of actions and then to break down those we encounter into their components.

7. Compare this concept to shooting the Archduke Ferdinand; this is not an institutionalized way of expressing individuality and is not, therefore, a formative action.

|                       | Open Context                                                       | Determinate Context                                             |
|-----------------------|--------------------------------------------------------------------|-----------------------------------------------------------------|
| Open Behaviour        | <b>OPEN<br/>POLIMORPHIC</b><br><br>Love<br>Letter<br>Writing       | <b>PLAYFUL<br/>POLIMORPHIC</b><br><br>Bored<br>Assembly<br>Line |
| Determinate Behaviour | <b>OCCASIONED<br/>POLIMORPHIC</b><br><br>Voting<br>with<br>Buttons | <b>MIMEOMORPHIC</b><br><br>Golf<br>Swing                        |

Figure 3.1  
Polimorphic actions and mimeomorphic actions.

(or, as we will come to see, by the envelope of our indifference to exactly how we carry out an action).

We call such actions *mimeomorphic* because the behaviors associated with the action can be copied from previous instances. Whereas polimorphic actions are both many-shaped and take their shape from society, mimeomorphic actions are same-shaped and take their shape from previous examples of the same action.

Four types of mimeomorphic actions can be assembled by combining two of their properties. Every mimeomorphic action must be more or less *special* or *casual* and it must be *singular* or *disjunctive*. The four types of mimeomorphic actions comprise the four combinations of these

two dichotomies though the casual–special dimension is more of a continuum than a dichotomy.

### **Special Singular Mimeomorphic Actions: Swinging a Golf Club**

When one swings a golf club—let us say one is hitting balls at a driving range—one seeks to produce (in most cases) a specific and fixed bodily movement every time. One puts one’s feet at a certain distance from one another, bends one’s knees, keeps one’s left arm straight, holds the club with one’s hands overlapping in a certain way, lowers one’s head such that one’s eyes remain fixed on the ball throughout the swinging movement, and so forth. The club is lifted into the air and swung back as the upper body turns, again in a specific and fixed fashion. The downswing starts with a sideways movement of the knees. The learner is told how to position and move almost every part of the body, from heels and toes to the top of the head, the aim being to develop the ability, first to execute these movements optimally, and then to reproduce the optimum movements time after time. As golfers know, it is hard to learn to produce the optimum movements even once, but it is still harder to reproduce them reliably. Even the best golfers in the world spend much of their time with their coaches adjusting and “grooving” their swing when it drifts away from the optimal.

In the modern game the same philosophy extends through all the golf clubs (except the putter) that are used to strike the ball. The golfer is advised not to try to hit the ball more softly when the target is at a lesser distance, but instead to choose a shorter, less powerful club, which, when swung in the same way, will have the effect of projecting the ball less far. Thus, for instance, a 7-iron in the hands of a good player hits the ball 150 yards and an 8-iron 140 yards. The experienced player who has mastered this standardized way of hitting the ball has more difficulty with shots that are “between clubs”—say 145 yards in length—than with a normal shot. To send the ball this distance, other things being equal, the golfer will have to hit unusually hard or unusually soft. Professional golfers who have played a bad shot in televised tournament often explain by saying that the shot was “between clubs.” It is worth noting that this problem does not arise for the poor player, who is not yet able to hit



shots in a mimeomorphic way. For the poor player a 150 yard shot is just as hard as a 145 yard shot.

What we have so far said about golf applies to today's game played according to today's coaching conventions. This coaching philosophy would lead the player to carry a separate club for each distance. It has not always been thus, as the following quotation illustrates:

Golfers weren't restricted to 14 clubs until the late 1930s. In any case, in earlier years, no one even thought of carrying such a vast quantity. Open Champions around the turn of the century regarded 7 to 10 clubs as the norm. Between the wars, all this changed dramatically. Rightly or wrongly, top professionals decided that it is far more difficult to be consistent when you have to vary your swing pace, and alter your clubface position at impact by manipulating it with the hands. They felt it was far easier to repeat a standard swing. Some of them began to carry, not 14 clubs, but quantities soaring into the 20's. (Hobbs 1991, p. 59)

It is clear, then, that before the interwar period swinging a golf club was not the sort of mimeomorphic action it is now; golfers were not trying to "groove their swings" in the same way.<sup>8</sup> One might say that the "morphicity" of the golf swing has changed.<sup>9</sup>

### **Special Disjunctive Mimeomorphic Actions: Expert Golf Club Swinging**

As golfers become more expert, they learn to reduce the variations in bodily movement as far as possible but, once they have mastered this, they may learn certain specific variations of the swing. For example, slight variations of position of hands and feet can result in side spin being imparted to the ball. The novice golfer tries desperately to eliminate unwanted side spin, such as gives rise to the dreaded "slice," by fixing the position of hands and feet, but the expert may introduce deliberate variations of stance to produce controlled spin so as to fly the ball around trees or away from dangers. Let us say that middle-ranking golfers can

8. They may have thought of their swings as a very complicated disjunctive mimeomorphic actions.

9. Jenkins (1994) has brought the analysis of the golf swing up to date. He finds that there are golf swing "paradigms" and paradigm revolutions. At one time what was generally thought to be the proper swing was modeled on that of Ben Hogan, at a later date on that of Jack Nicklaus, and more recently on Nick Faldo's swing.

hit three types of shot: the straight shot, the left to right shot, and the right to left shot. Imagine such a golfer standing with a tree between him- or herself and the target. If the tree is tall and the bulk of it is to the golfer's left, then he or she will go around the right-hand side by hitting a right to left shot. If the tree is tall but the bulk is to the golfer's right, then a left to right shot is called for. If the tree is short, then he or she will hit a straight shot over the top. Where the circumstances are such that the choice can be predicted, this is a *disjunctive mimeomorphic action*. We call it "disjunctive" because the actor has to choose to execute the action in one or other of a set of mutually exclusive ways, each of which is itself a mimeomorphic action.

Whether to describe this three-choice golf swing as a disjunctive mimeomorphic action or three individual singular mimeomorphic actions is a matter for the analyst; both descriptions are correct. Because actions are embedded within other actions, there are always choices of this kind to be made by the analyst. All *disjunctive mimeomorphic actions* could be described as combinations of individual *singular mimeomorphic actions*.

It is important to note the difference between disjunctive mimeomorphic actions and *occasional polymorphic actions* (our example was voting with buttons). In this case, as we have just said, three separate singular mimeomorphic actions can be thought of as one combined disjunctive mimeomorphic action. In the voting case discussed above, a button pressing, considered on its own, is a mimeomorphic action. But the three possible mimeomorphic actions that make up the voter's choice do not constitute a disjunctive mimeomorphic action—they constitute a polymorphic action, because the choice between them is socially situated. In the golf case, the choice is predictable from the outside. Assuming the golfer is trying to do his or her best in terms of the game as it is currently played, rather than, say, "showing off," the shot can in principle be predicted from the relative position of the ball and the tree. That one is predictable and the other is not has to do with the openness of the context in the case of voting and the closedness in the case of our golf example, where the three choices of shot correspond to three well-defined and specifiable situations.

Another example of a disjunctive mimeomorphic action is chair spraying. Imagine a Taylorist assembly line in which workers are trained to spray chairs in the most efficient manner for each shape of chair. Let there be three models of chair passing along the assembly line in random order. The movements that the chair sprayer makes will be one of three mimeomorphic actions, the relevant mimeomorphic action being chosen to match the model of chair passing at that moment. The universe of chairs is exhaustive—there are no in-between cases—and the sprayer's behavior would be predictable.

But what if there are in-between cases? The problem for the golfer is easily seen. A really good golfer might break down the “obstructing-tree-situation” into many more than three cases. In fact, for a really good golfer there is a “continuous” scale of possibilities. Such a golfer would try to put more or less side spin on the ball, depending on just how far the tree obtrudes into the line of the shot, and therefore on how far the flight path of the ball has to be “bent.” The really good golfer's rule about how much spin he or she would like to put on the ball can be expressed as a function of the extent to which the tree protrudes into the line of the straight shot. This is still mimeomorphic action because in every case the degree of spin preferred by the golfer could be predicted in advance, even though every case cannot be anticipated; what can be anticipated is the full range of possibilities and the full range of eventualities that would lead to the choice of a shot from the range.<sup>10</sup>

The golfer, choosing from a range of degrees of side spin according to the extent of intrusion of the tree into the proposed flight of the ball, is responding to the environment (through visually mediated “feedback”). A more obvious case of feedback is, say, a person polishing a panel of a car. The person keeps the polishing rag in contact with the surface of the

10. In Collins (1990) a “record-and-playback” chair sprayer was discussed. The initial recording of the human chair sprayer's movements is a matter of recording a continuous function. Furthermore, the example allows for any number of different chair shapes to be recorded and played back. As the number becomes large, the difference between discrete and continuous rules gets lost. To put this another way, no applicable function is truly continuous, because the world is not truly continuous.

paintwork by pressing down and allowing the shape of the metal to guide the muscles. This is still *disjunctive mimeomorphic action*, because, once more, the range is understood in advance and the function that accounts for the response to the environment is understood in advance. In this case the full set of movements that might be described cannot be predicted in advance because one cannot predict the shape of the panels of cars yet to be designed. Nevertheless, one can predict the envelope of shapes and thus the function. Or, to put this the other way, it is only to the extent that one cannot predict the range of shapes that one cannot say that car polishing is mimeomorphic. If someone starts designing car panels with sharp-edged troughs or spikes, normal car polishing will break down, and something new will have to be invented.

### Casual Singular Mimeomorphic Actions: Telephoning One's Mother from Home

To telephone one's mother from home, one must engage in a mimeomorphic action. If one has, say, a push-button phone, one must pick up the handset and press the same sequence of buttons in the same order every time one telephones. Therefore one is trying to execute that action—telephoning mother—with the same behavior on each occasion. Notice, however, that this action is different from swinging a golf club, in that we are not trying to standardize the position of other parts of our body. We do not need to keep a stiff left arm; we do not need to bend our knees or place our feet at shoulder width; we do not need to keep our head still; we do not need to grip the phone in just such and such a way; nor need we press the buttons with just such and such a pressure or tempo. While there are limits outside of which our movements must not stray, the envelope is much wider than in the case of golf. We do not need to concentrate or undergo lengthy training in order to remain within the limits of telephone manipulation that count as skilled, nor do we care whether we press the buttons and hold the phone in just the same way on different occasions or in countless different ways. In short, whereas in the case of the golf swing our attention is all on the matter of standardizing our movements as far as possible, in the case of telephone dialing we are *indifferent* to the range of movements that lies within the

envelope of acceptable behavior—that is, the range of movements that will amount to calling up the relevant number.

This is a *casual action*, as opposed to a *special mimeomorphic action*, because the mimeomorphic character of the action is not marked, as in the case of a special mimeomorphic action, by our preference for the same behavioral instantiation each time, but by our indifference to how the action is behaviorally accomplished (within limits). This indifference is made evident by the following observation: We would not care if the action was carried out every time with identical bodily movements. It is for this reason that it makes sense to call such actions mimeomorphic even though our attention is not focused on maximizing identity of behavior. The contrast with polymorphic actions is still clear: Both polymorphic actions and *casual mimeomorphic actions* involve variety in the behavioral instantiation of actions, but in polymorphic actions the variations are meaningful and cannot be collapsed into the same behaviors every time, whereas in a casual mimeomorphic action they are not meaningful and they can be collapsed without loss. To put this another way, as opposed to the case of polymorphic actions, the variability with which casual mimeomorphic actions can be realized is *not* a part of what the action means for the agent.

As we will go on to see, even special mimeomorphic actions have an element of casual mimeomorphic action about them, so that the difference is one of degree, though the difference in intention is often very clear, as in the telephone and golf swing cases. Degrees of variability in the behavioral instantiation of an action exist, as it were, phenomenologically: they exist for the agent and they are an objective feature of the agent's community, but they do not map straightforwardly onto degrees of variation in the physically observable features of the action.

### Casual Disjunctive Mimeomorphic Actions: Telephoning

In the case of telephoning one's mother from home, one is always using the same phone and always calling the same number. In the case of telephoning in general, one calls from a variety of phones—pay phones with buttons, phones with dials—and calls a variety of numbers. Each action of telephoning is a mimeomorphic action that is designed for the

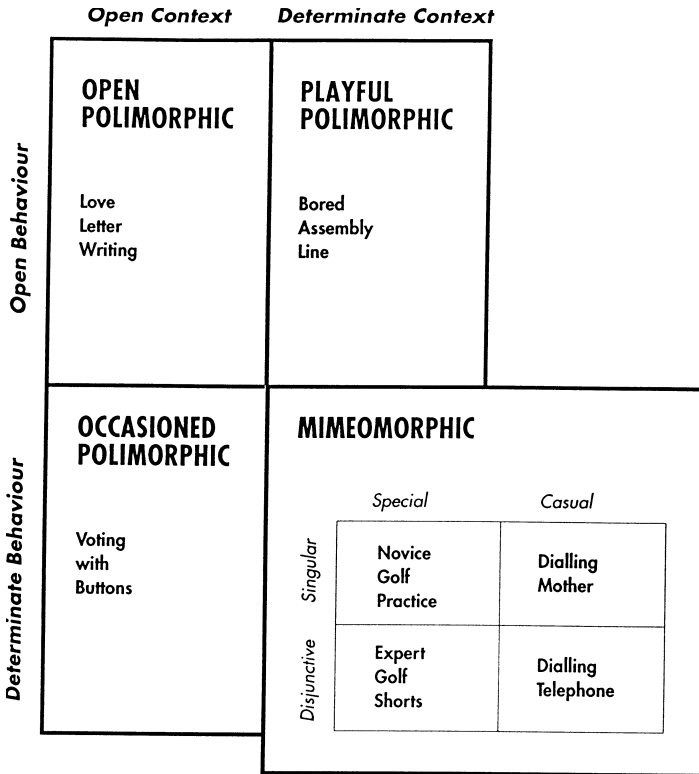
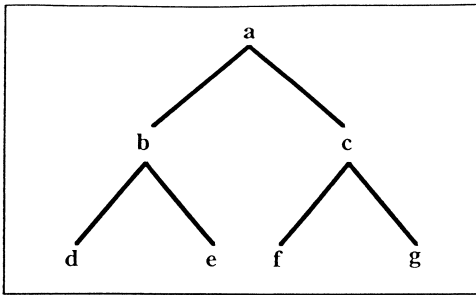


Figure 3.2  
Seven types of actions and their relationship.

particular phone and for the particular number. For each of our interactions with telephones, what we said about telephoning one's mother is true, but each case is fitted to the particular context which determines, in a predictable way, the envelope of behaviors that make up the mimeomorphic action. Once more, as in the expert golf swing or the chair-spraying example, a set of mimeomorphic actions—in this case a set of telephonings—is assembled to form a higher level disjunctive mimeomorphic action. In this case, however, the higher level choice is between casual mimeomorphic actions, so this is an example of a casual disjunctive mimeomorphic action.

All seven types of morphicity of actions are set out in figure 3.2.



**Figure 3.3**  
Traveling to London by train.

### Diagrammatic Representation of Actions

Many relationships between actions are better understood and more easy to analyze with the use of diagrams. We use open and closed triangles to represent actions, the width of the base representing the degree of freedom of the behavioral instantiation.

#### Polimorphic Actions: Open Triangles

In what we will call an *action tree*, higher nodes stand for types of actions that are realized lower down the tree by more precisely defined action types. Thus for instance “traveling to London on the early morning train” (labeled as “d” in figure 3.3), falls under “traveling to London by train” “b,” which in turn falls under “traveling to London” “a.”

Of course the tree structure is no more then an idealization of relations between types of actions, since a given action can often be specified in more than just two ways (e.g., writing a love letter can be specified in an endless number of ways), so the bifurcation of figure 3.3 is too simple. What is more, there may be many more levels to the tree. We will analyze trees, and the paths through them that actors adopt, in greater detail below. In the meantime we will introduce a diagrammatic shorthand for the complexities of an action tree. In this convention, we collapse all the levels and represent the open-ended branching of the tree by an open triangle. Thus a given action type with its possible realizations will be represented as in figure 3.4. Here, “a” stands for a type of action such

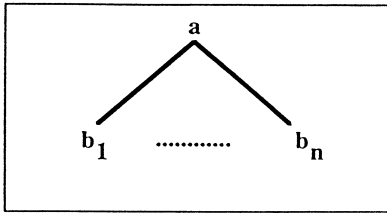


Figure 3.4  
Collapsed action tree.

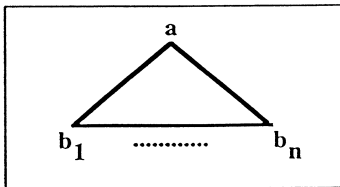


Figure 3.5  
Mimeomorphic action represented by a closed triangle.

that it can be carried out in an indefinite number of ways and such that the ways “b1” to “bn” in which the action “a” can be carried out, are significant for the actor and/or his or her community. The actor chooses one way of carrying out action “a” on a given occasion, that is, selects one possibility from the *domain of choice* “b1” to “bn.”

### Mimeomorphic Action: Closed Triangles

The actions of swinging a golf club (on a driving range without trees in the way), or Taylorist chair spraying do not have open action trees. To begin to develop a representation, we take a casual singular mimeomorphic action such as telephoning one’s mother, symbolized by a closed triangle as in figure 3.5. Here “a” stands for a type of action such that actors (or their community, or both) *do not care about* which way, “b1” to “bn,” they carry out action “a,” so long as it falls within the triangular envelope of possibilities. The actor happens to carry out the action “a” in some way on a given occasion, but takes no interest in which particular element from “b1” to “bn” was used, or in varying the instantiations of this action. Thus in this case, “b1” to “bn” do not constitute an open



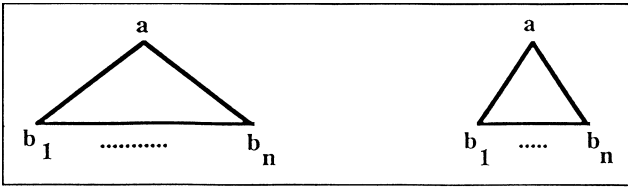


Figure 3.6

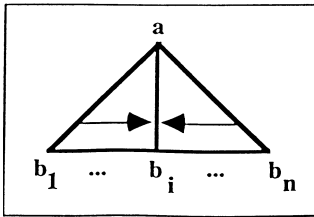
Mimeomorphic actions with different envelopes of behavior.

domain of choice, but rather an *area of tolerance* or *indifference* to which the actor does not attend.

### Casual and Special Mimeomorphic Actions: Width of Triangles

Moving downward in an action tree amounts to fixing more and more aspects of how the higher action type is to be carried out on a given occasion. At the branching nodes, choices must be made, either explicitly and consciously, or else implicitly and unconsciously. Put differently, we might say that the downward movement fixes more and more parameters of an action. Focusing for the moment only on mimeomorphic actions, let us imagine ourselves constructing a rough and ready list of the things that are important in our examples of *special* and *casual mimeomorphic actions*, golf club swinging and telephoning. In the case of the golf swing, the list would include almost every part of our body, and the specification for the movement of each of these parts of our body would be narrow. In the case of telephoning, a few movements of the arms and hands would be specified within wide limits. The degree of constraint can be represented by narrow and wide triangles, as in figure 3.6. The more parameters of a mimeomorphic action are fixed—the smaller the area of tolerance or indifference—the narrower the triangle.

This kind of representation helps us to understand the interplay of the intentional description of action and the physical description. In the case of special mimeomorphic actions, of which the learner's golf swing is a paradigm, the actor is attempting to collapse the triangle entirely—to eliminate variation altogether. And this is what it feels like! From the point of view of an actor, then, a special mimeomorphic action is an attempt to collapse the triangle into a straight line, as in figure 3.7. But, described in terms of what an outside observer would see of our rough

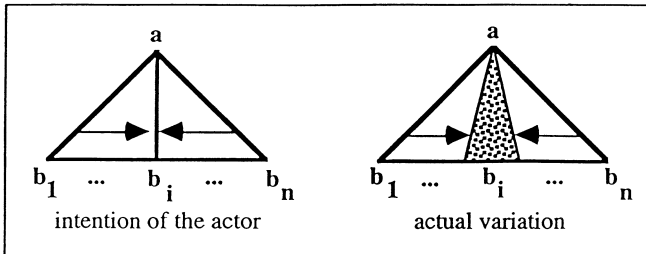


**Figure 3.7**  
Ideal type of a special mimeomorphic action.

list of physical and physiological parameters, the learner's triangle would be far more open because learners are not very successful at collapsing the triangle. For an outside observer, the degree of physical variation in the swing of a learner, and in the swing of a professional "clowning about" and imitating a learner, would be similar, but the intentions would be quite different. In the first case, the learner would be engaged in special mimeomorphic actions (though not very successfully); in the second case the professional would be engaged in the action "clowning about," which is polymorphic action.<sup>11</sup> Thus in terms of physical parameters, two actions that fall on the opposite sides of our major dichotomy look the same; once more we see the difficulty of doing social science "from the outside" unless there is nothing going on but mimeomorphic action.

Now suppose the learner succeeds and becomes a professional who is having a good day on the practice ground, so that as far as he or she, or a passing observer, or a watching golfing coach is concerned, the triangle has been collapsed. Even then, the triangle is still open within other frames of reference. As we remarked earlier, within the framework of the person whose dog is hit by one shot but not by another, or within the framework of an engineer with a high-speed camera and a micrometer, two golf swings that seem identical to the golfer will look different. Thus if one attends even to successfully executed special mimeomorphic actions within the right frame of reference, one will see that the straight line is really a narrow triangle, as in figure 3.8. Once again, that it is possible

11. Probably it is a playful polymorphic action because the variations in the behavioral instantiation are deliberately opened up, but the variations would be nearer to random than meaningful.



**Figure 3.8**

Close-up view of a special mimeomorphic action.

for an actor from one community to see two actions as the same, whereas an actor from another community would not, is a matter of the respective areas of tolerance that are used to define sameness.

To sum up, in the case of special mimeomorphic actions, the actor is attempting to collapse the triangle, while the actor in casual mimeomorphic actions is indifferent to what happens within the base of the triangle. Should the triangle collapse, and the action be carried out in what might appear to be the identical way every time, the actor would not care. (We would not care if, every time we phoned our mothers, we had our feet, knees, head, etc., in exactly the same position, or whether we always pressed the buttons with the same force and tempo. We would not even notice should this be the case.) It might be said, then, that special mimeomorphic actions and casual mimeomorphic actions merge into one another, for even in special mimeomorphic actions the actor is indifferent to variations that do not belong to the frame of reference of the activity in question (the golfer is indifferent to the differences noticed by the dog owner and the engineer.) As we will see, the similarities between all kinds of mimeomorphic actions are, for many purposes, more important than the differences; the differences, however, reflect our experience as actors and observers.

### **Playful Polimorphic Actions Revisited**

At this point it is worth returning to *playful polimorphic actions*. Playful polimorphic actions occupy an interestingly ambiguous position within the scheme; this makes the significance of the major division still more clear. We have located playful polimorphic actions on the polimorphic

action side of the major dichotomy because the actor would not be indifferent to a collapse of the triangle into a straight line. The unusual thing is that the actor's concern is not with the meaningfulness of the various positions within the triangle that represent different behavioral instantiations of the action, but solely with keeping the triangle opened up—to relieve tedium. The only meaningfulness is in the width of the triangle, not the structure within it; in this, playful polymorphic actions are unlike the other two categories of polymorphic actions. The meaningfulness of the internal structure of playful polymorphic actions makes them mimeomorphic actions more than polymorphic actions. That is that is why they are an interesting interstitial case.

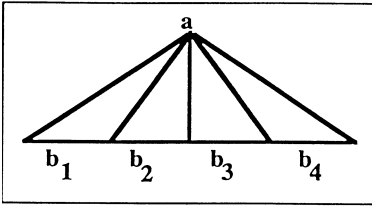
Compare playful polymorphic actions with occasioned polymorphic actions (e.g., voting with buttons). In the latter case, the width of the triangle is very narrow in terms of most frames of reference. Thus in our voting example only one of three choices is available. Nevertheless, the meaning of the structure within that narrow triangle of choices is crucial.

### **Disjunctive Mimeomorphic Actions: Divided Triangles**

In *disjunctive mimeomorphic actions* the behavior is related to a determinate set of contexts such that one mimeomorphic action is required for each prespecified (or potentially prespecifiable), situation.<sup>12</sup> Our clearest example of this was chair spraying on a Taylorist assembly line. We represent disjunctive mimeomorphic actions as a closed triangle divided into sections as in figure 3.9. Each line of the triangle represents one of the ways in which the action will be executed.

One might say that the research program of artificial intelligence (at least, the program that preceded neural nets and so forth), involves

12. "Potentially prespecifiable" includes cases where the actor (or delegate) is provided with an algorithm for recognizing the situation even though the exact constituents of the situation may not have been thought through in advance. For example, an algorithm of this sort is used by skilled people who have to drill holes in plates. Skilled hole drillers know that the torque experienced by a drill bit increases just before the bit breaks completely through. (When the residue of the material being drilled becomes so thin that the forward pressure of the drill distorts the material, the torque will momentarily increase.) They know they should relieve pressure on the drill at this moment. This feature of drilling is used in the design of machines to drill holes automatically in surgical settings such as automated ear surgery. (Annemiek Nelis provided this example.)



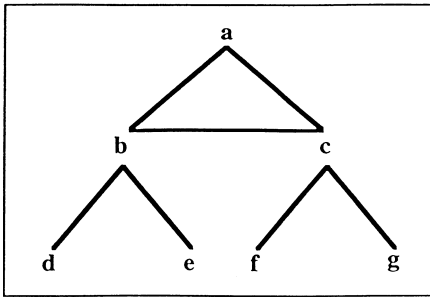
**Figure 3.9**  
Disjunctive mimeomorphic action.

making machines that do the behavioral equivalent of more and more ramified trees of disjunctive mimeomorphic actions. Those who believe in “strong AI” think that all actions are ramified trees of disjunctive mimeomorphic actions. We would say that such people confuse densely ramified *disjunctive mimeomorphic actions* with polymorphic actions. They think that as a triangle becomes more and more densely filled with alternate paths, it becomes indistinguishable from an open triangle.<sup>13</sup>

### Actions Embedded within One Another

Every mimeomorphic action in part realizes a higher action type that need not itself be a mimeomorphic action; in other words, as we have already mentioned, mimeomorphic actions can be embedded in polymorphic actions. For instance, swinging a golf club in part realizes the higher action type “playing golf,” an action that is not mimeomorphic. Playing golf includes actions such as negotiating with the caddie, estimating distances, judging wind conditions, considering one’s opponent’s strengths and weaknesses, “gamesmanship,” and so forth. All of these are polymorphic actions (though they may each contain components that are mimeomorphic actions). If the sequence of realizations is followed high enough, all actions are embedded in polymorphic actions, because all actions are embedded within societies.

13. Some of the arguments about the Turing test are about the distinguishability or otherwise of these two kinds of actions as they reveal themselves in conversation. The person in John Searle’s “Chinese Room” is engaged in pure *disjunctive mimeomorphic actions*. These are distinguishable from polymorphic actions if the appropriate questions are asked (Collins 1990, chaps. 13 and 14). See also Collins (1996), where the Turing test is replaced with the “editing test.”



**Figure 3.10**  
An impossible action tree.

Though mimeomorphic actions can be embedded in polymorphic actions, polymorphic actions cannot be embedded in mimeomorphic actions. In terms of our diagrammatic representation, figure 3.10 is impossible. For action “a” to be a mimeomorphic action, the agent should be indifferent to how “a” is carried out in terms of behavior, as long as it falls within the area of tolerance. And this can only mean that the indifference should go all the way down, that is, “b” and “c” should be closed as well. If “b” or “c” were open, the way that “b” or “c” was carried out would be a matter of concern to the actor, and this would mean that the way “a” was executed would be of concern, because “a” is executed through the execution of “b” or “c.”

As we said, one can have mimeomorphic actions within polymorphic actions. This is because the variations of behavior through which the action is executed are given their contextual meaning at the higher level of the embedding action.

Nearly all polymorphic action trees have mimeomorphic action embedded within them. In many cases, as one goes down an action tree, the ratio of mimeomorphic actions to polymorphic actions will increase, and nearly all action trees will terminate in “mimeomorphic actions.”<sup>14</sup> Thus,

14. To anticipate our major argument, because mimeomorphic actions can be delegated to entities that do not understand contexts but will work according to plans, the lower levels of action trees can often be delegated to tools. A frequent mistake is to try to delegate too far up the action tree. To put this another way, the mistake is to shift the locus of control from the human tool user to the tool.

nearly all actions, including polymorphic actions, are, as it were, assembled out of mimeomorphic actions. An assembly of mimeomorphic actions might, then, constitute either a polymorphic action or a disjunctive mimeomorphic action; it depends on the kind of choice that is being made at the higher level. Note that this does not mean that a polymorphic action can be reduced to an assembly of mimeomorphic actions; it is the way the mimeomorphic actions are arranged and chosen that constitutes a polymorphic action.<sup>15</sup>

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See Lipscombe (1989) for an analysis of this problem in the context of medical expert systems.

To save possible confusion, consider the following. Suppose a naive observer watches a virtuoso piano player. He or she sees only a sequence of key presses. The observer might reproduce these in a mimeomorphic way, believing that he or she has reproduced the action. The reproduced action, however, would be characterized by the novice's indifference to the exact way each key was pressed, since he or she will think that this was also a matter of indifference to the pianist. Virtuosi, or skilled critics, however, would be able to see the difference. The naive observer would have believed they were watching an action that could be described "externally," as one might say. The virtuoso or the skilled critic would know that this was not so. But replace the novice observer with a device for recording and playing back the piano key pressings "exactly" as they were played—for example, think of a pianola roll being played back on a modern recording piano. Surely then the polymorphic action is being reproduced by a device that cannot respond to social context! But the virtuoso pianist must play according to the response of the audience and the mood of the times; if not, there would be no need for live concerts. A musical performance is like a love letter (see below). To replay a pianola roll is like rereading a love letter. To substitute the roll for the live performance is like sending the same letter over and over again.

This way of putting the matter shows how the role of piano playing might correspond to the role of language in a Turing-like test of the pianist. Novice piano playing, like novice (Chinese Room) language speaking, is generally mimeomorphic, whereas virtuoso piano playing is not. (But not all expert activity is polymorphic—see chapter 5).

15. There is an interesting comparison to be made with ideas such as that contained in Minsky's (1987) "society of mind." The socially embedded feature of polymorphic actions is *not reduced* by analysis into mimeomorphic elements. Higher-level polymorphic actions can be executed only by properly socially embedded actors. Entities that reproduce the behaviors corresponding to the lowest level of an action tree would not be able to match the choice of behaviors to social context that is required if a polymorphic action is to be successfully executed.

The reason nearly all actions terminate in mimeomorphic actions is that as we go down the tree structure, we reach ever more precisely specified action types, and almost inevitably we cease to care how these already highly specified actions are further specified. We cannot care about the exact way we carry out the most minute elements of our actions because humans' ability to attend to detail is not that great.

This is not to say that the bodily movements to which we never attend are found at a fixed level. There are bodily movements that we do not normally attend to, which are "things we do in order to do actions rather than being actions themselves"; but at different times and places even these might turn into actions. For example, though it is nearly always the case that when our heart is beating it is making it possible for us to do other actions rather than comprising an action itself, the invention of "biofeedback technology" carries with it new possibilities. Perhaps we could turn the working of the heart muscles into an action, narrowly conceived; this in turn carries with it the possibility of heartbeating becoming a formative action.

We have to say that *nearly* all actions terminate in mimeomorphic actions, because there seem to be cases where it would be petty to press the point. For example, the action of hand-writing a letter, rather than using a typewriter, is best thought of as terminating in a playful polymorphic action of the personalized variety; the inscriptions of the pen are the last thing we attend to, and we try to vary these. But playful mimeomorphic actions are, as we have explained, a hybrid category. Perhaps it is the case that all action trees terminate in actions in which the exact specification of the behavior is not meaningful—they terminate, then, either in mimeomorphic actions or in playful polymorphic actions. We will take this to be the case.

Putting together the last few paragraphs, we find that if we extend actions trees far enough they are always polymorphic at the top and (nearly) always mimeomorphic at the bottom. When we say an action is a polymorphic action, we know it will almost always have mimeomorphic actions embedded in it. Conversely, when we say an action is mimeomorphic, we know that it will be embedded in a polymorphic action.